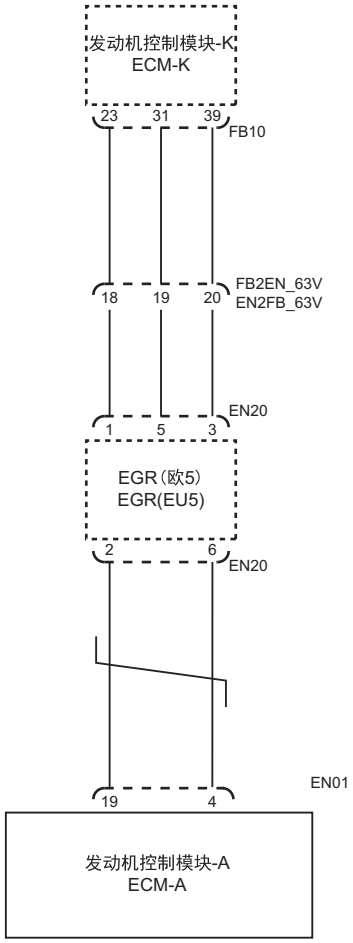
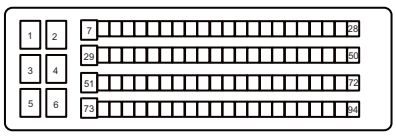
P213A、 P0487

Description of fault diagnosis codes
P213A-01：Short circuit over load error for H-Bridge EGRVlv powerstage
P213A-11：Short circuit to ground on EGR H-bridge Out2 error
P213A-123：Short circuit to battery on EGR H-bridge Out2 error
P213A-13：Open load error for EGRVlv H-Bridge powerstage
P213A-19：Temperature dependent over current error for EGRVlv H-Bridge powerstage
P2621-11：Over temperature error for EGRVlv H-Bridge powerstage
P2621-11：Over current error for EGRVlv H-Bridge
P2622-12：Under voltage error for EGRVlv H-Bridge powerstage
P0487-11：Short circuit to ground on EGR H-bridge Out1 error
P0487-12：Short circuit to battery on EGR H-bridge Out1 error
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Fault in harness or harness plug • EGR valve fault
Description of circuit diagram
Exhaust recirculation valve is a mechanical-electrical integrated product installed on the diesel engine, which is used to control the exhaust recirculation fed back to the air intake system, the engine control module provides 12v voltage to the drive motor and also provides grounding to the low reference voltage.

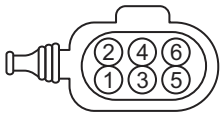
Circuit diagram



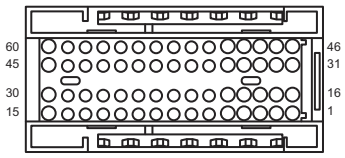
FB10



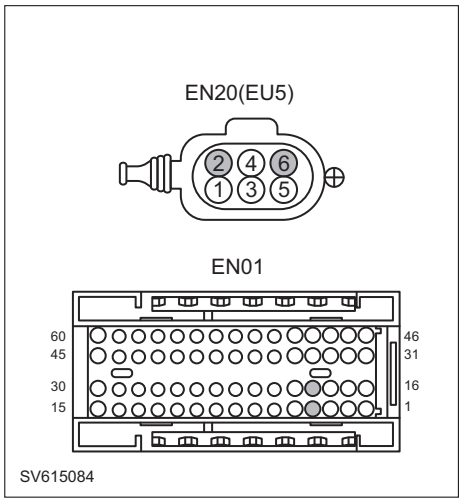
EN20(EU5)



EN01



SV615011

Troubleshooting procedures		
Step 1	General inspection.	
		1 Check the fuse for damage. 2 Check battery capacity. 3 Check the engine control module and harness plug of EGR valve for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not.
		No → Repair or replace the faulted parts.
No ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the engine control system with diagnostic apparatus. 4 Is there any other fault code except for P213A, P0487?
		Yes → Reference:DTC Summary List (EU5).
No ↓		
Step 3	Check the driving circuit of EGR valve (open circuit).	
		1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of EGR valve. 3 Disconnect the harness plug of engine control module. 4 Measure the resistance of harness plug between EGR valve and engine control module. Measuring circuit: the resistance between EN20 plug terminal 2 and EN01 plug terminal 19. Measuring circuit: the resistance between EN20 plug terminal 6 and EN01 plug terminal 04. **Standard value: ≈ 0Ω** 5 Check whether the result is normal?
		No → Repair or replace the faulted parts.
Yes ↓		

Step 4	Check the circuit of EGR valve (short circuit).
	1 Turn the ignition switch to OFF. 2 Measure the resistance between the harness plug of EGR valve and ground. Measure circuit: the resistance between EN20 plug terminal 2 and ground. Measure circuit: the resistance between EN20 plug terminal 6 and ground. **Standard value: ∞** 3 Measure the voltage between the harness plug of EGR valve and ground. Measure circuit: the voltage between EN20 plug terminal 2 and ground. Measure circuit: the voltage between EN20 plug terminal 6 and ground. **Standard value: 0V** 4 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	
Step 5	Check the driving circuit of throttle valve (components).
	1 Measure the resistance of EGR valve terminal. Measure circuit: the resistance between EGR valve terminal 2 and terminal 6. **Standard value: 14~19Ω** 2 Check whether the result is normal?
	No → Replace the EGR valve.
Next ↓	
Step 6	Complete the check and confirm whether the system works normally nor not.